The Fixed-Target Theorem

Jake Brukhman
jake@coinfund.io

GPT 5.6 Sol

Abstract

Seymour's Second Neighborhood Conjecture asserts that every finite oriented graph has a vertex with at least as many exact second outneighbors as outneighbors. Established cases include tournaments, proved by Fisher [1], and oriented graphs of minimum outdegree at most six, proved by Kaneko and Locke [2]. We give a short fixed-target counting proof that the conjecture holds for every oriented graph of order n and minimum outdegree δ satisfying $n \leq 2\delta + 2$.

1 Definitions and statement

A finite *directed graph* is a pair $G = (V(G), A(G))$, where $V(G)$ is a finite set of vertices and $A(G)$ is a set of ordered pairs of distinct vertices, called *arcs*. We write

$$u \to v$$

when $(u, v) \in A(G)$. Thus the arrow points from the source of the arc to its target. The directed graph is *oriented* if, for every two distinct vertices u, v, at most one of $u \to v$ and $v \to u$ is present.

For a vertex $v \in V(G)$, define its outdegree and indegree by

$$d^+(v) = |\{x \in V(G) : v \to x\}|, \qquad d^-(v) = |\{x \in V(G) : x \to v\}|.$$

The minimum outdegree of G is

$$\delta = \min_{v \in V(G)} d^+(v).$$

For a vertex $v \in V(G)$, define the exact first outneighborhood by

$$N_1(v) = \{x \in V(G) : v \to x\},$$

and the exact second outneighborhood by

$$N_2(v) = \{x \in V(G) \setminus N_1(v) : y \to x \text{ for some } y \in N_1(v)\}.$$

Direct outneighbors are excluded explicitly. The vertex v is excluded automatically: a path $v \to y \to v$ would be a directed 2-cycle, which an oriented graph cannot contain. Thus $N_2(v)$ consists exactly of the vertices reached from v in two steps but not in one step. In a reachability statement from v to x, we call v the *source vertex* and x the *target vertex.*

A vertex v is a *Seymour vertex* if

$$|N_2(v)| \geq |N_1(v)|.$$

Seymour's Second Neighborhood Conjecture says that every finite oriented graph has a Seymour vertex.

Theorem 1 (Fixed-target theorem)**.** *Let G be an oriented graph on n vertices with minimum outdegree δ. If*

$$n \leq 2\delta + 2,$$

then G has a Seymour vertex.

2 The fixed-target capacity bound

For a source vertex v, define

$$U(v) = V(G) \setminus \big(\{v\} \cup N_1(v) \cup N_2(v)\big).$$

Thus $U(v)$ is the set of target vertices that the source v cannot reach in at most two steps.

Now fix a target vertex x. Define the *trap* of x by

$$T(x) = \{y \in V(G) \setminus \{x\} : y \not\to x\}.$$

These are exactly the vertices that do not send an arc to x. Because G is oriented, every outneighbor of x belongs to $T(x)$. Hence

$$|T(x)| \geq d^+(x) \geq \delta.$$

Define the *target slack*

$$q(x) = |T(x)| - \delta \geq 0.$$

Here both $|T(x)|$ and δ count vertices: δ is the minimum number of outneighbors at any vertex. Thus $q(x)$ is the number of vertices by which the trap exceeds the size of a minimum outneighborhood.

Define

$$P(x) = \{v \in V(G) : x \in U(v)\}.$$

Thus $P(x)$ is the set of source vertices from which the fixed target x cannot be reached in at most two steps. If $v \in P(x)$, then $v \not\to x$. Moreover, no vertex of $N_1(v)$ points to x, since otherwise there would be a two-step path from v to x. Therefore

$$\{v\} \cup N_1(v) \subseteq T(x). \tag{1}$$

This is the fixed-target observation: every source missing x, together with all of its outneighbors, is confined to the same trap $T(x)$.

Lemma 2 (Fixed-target capacity)**.** *For every target vertex x,*

$$|P(x)| = 0 \quad \text{if } q(x) = 0, \qquad |P(x)| \leq 2q(x) - 1 \quad \text{if } P(x) \neq \varnothing.$$

Proof. By (1), every arc whose source lies in $P(x)$ has its target in $T(x)$. There are at least $\delta|P(x)|$ such arcs. At most $\binom{|P(x)|}{2}$ have both endpoints in $P(x)$, because G is oriented, and at most $|P(x)|\big(|T(x)| - |P(x)|\big)$ go from $P(x)$ to the rest of the trap. Consequently, if $P(x) \neq \varnothing$, then

$$\delta|P(x)| \leq \binom{|P(x)|}{2} + |P(x)|\big(\delta + q(x) - |P(x)|\big).$$

Dividing by $|P(x)|$ and rearranging gives $|P(x)| \leq 2q(x) - 1$. This is impossible when $q(x) = 0$, so then $P(x) = \varnothing$. □

The point of the lemma is that target slack is the only room available for packing many source vertices that all miss the same target.

3 Proof of the theorem

Suppose for contradiction that G has no Seymour vertex. Then $\delta > 0$, since a vertex of outdegree zero is automatically a Seymour vertex. Counting arcs by their sources gives

$$n\delta \leq |A(G)| \leq \binom{n}{2}.$$

Therefore $n \geq 2\delta + 1$. Together with the hypothesis, this leaves

$$c := n - 2\delta \in \{1, 2\}. \tag{2}$$

Let

$$L = \{v \in V(G) : d^+(v) = \delta\}, \qquad \ell = |L|.$$

Every $v \in L$ is not a Seymour vertex, so $|N_2(v)| \leq \delta - 1$. Hence

$$|U(v)| = n - 1 - \delta - |N_2(v)| \geq n - 2\delta = c. \tag{3}$$

Thus each minimum-outdegree source contributes at least c unreachable targets.
Let

$$I = \left|\{(v, x) \in V(G)^2 : x \in U(v)\}\right|.$$

Counting these ordered source–target pairs first by their source and then by their target gives

$$I = \sum_{v \in V(G)} |U(v)| = \sum_{x \in V(G)} |P(x)|. \tag{4}$$

This explains the double count: $x \in U(v)$ and $v \in P(x)$ describe the same ordered pair (v, x). From (3),

$$I \geq c\ell > 0. \tag{5}$$

Put

$$Q = \sum_{x \in V(G)} q(x).$$

Since $I > 0$, at least one set $P(x)$ is nonempty; the capacity lemma then implies that at least one $q(x)$ is positive. Applying the lemma in (4) gives

$$I \leq \sum_{q(x)>0} \big(2q(x) - 1\big) < 2 \sum_{x} q(x) = 2Q. \tag{6}$$

So the total target capacity is strictly less than twice the total target slack.
It remains to compare Q with ℓ. Since $|T(x)| = n - 1 - d^-(x)$,

$$q(x) = n - 1 - \delta - d^-(x).$$

Total indegree equals total outdegree, and every vertex outside L has outdegree at least $\delta + 1$. Therefore

$$\begin{aligned} Q &= n(n - 1 - \delta) - \sum_{x} d^+(x) \\ &\leq n(n - 1 - \delta) - \big(\ell\delta + (n - \ell)(\delta + 1)\big) \\ &= \ell + n(c - 2). \end{aligned} \tag{7}$$

This degree ledger bounds the total room in all target traps using only the number of minimum-outdegree vertices.
If $c = 1$, then (7) gives

$$0 < Q \leq \ell - n \leq 0,$$

a contradiction. If $c = 2$, then (5)–(7) give

$$I < 2Q \leq 2\ell \leq I,$$

again a contradiction. Hence G has a Seymour vertex.

References

[1] D. C. Fisher, *Squaring a tournament: A proof of Dean's conjecture*, Journal of Graph Theory **23** (1996), 43–48.

[2] Y. Kaneko and S. C. Locke, *The minimum degree approach for Paul Seymour's distance 2 conjecture*, Congressus Numerantium **148** (2001), 201–206.